

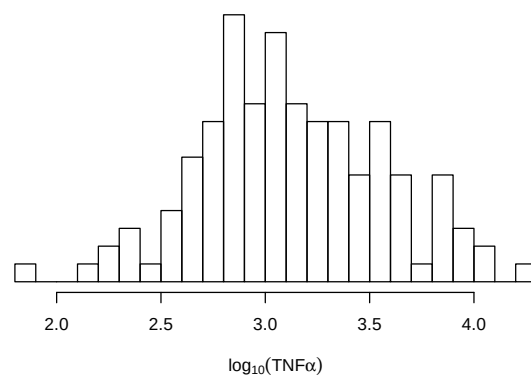
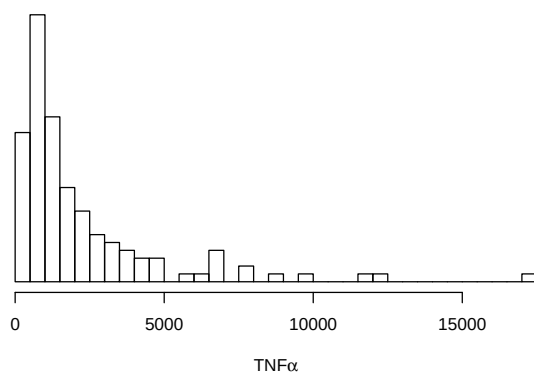
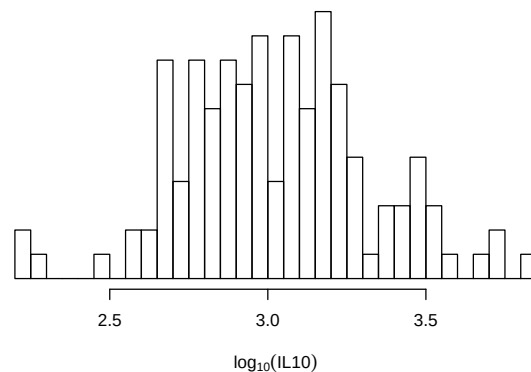
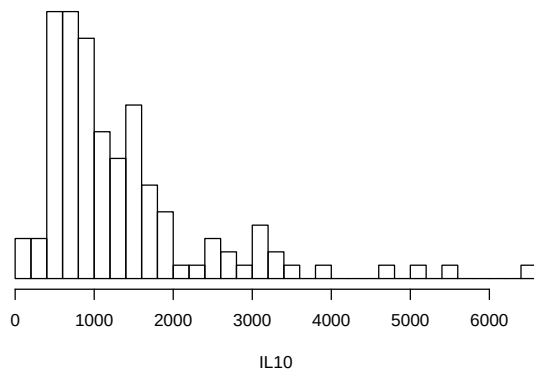
Possible transformations

- Logarithm
- Square root
- No transformation

Why transform?

- Obtain approximate normality
- Stabilize variation
- More informative graphs
- Obtain symmetry

Highly skewed data: take logs

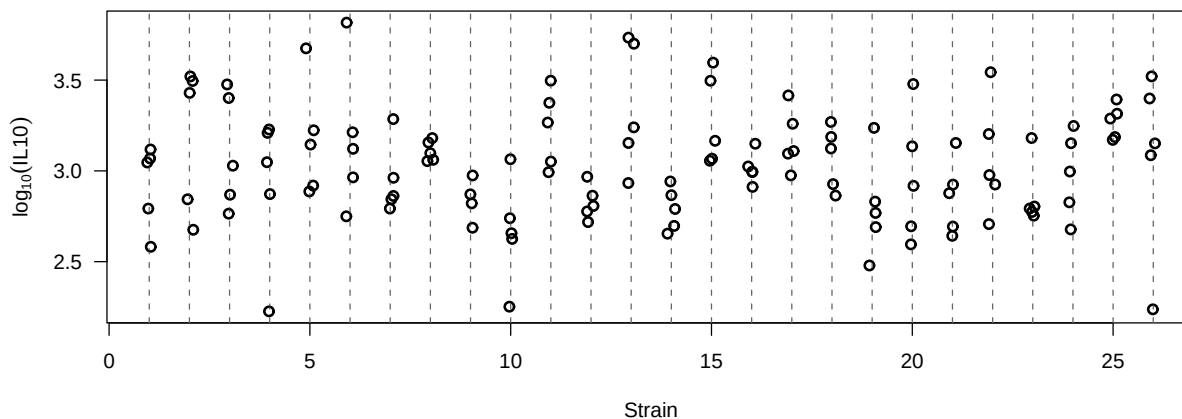
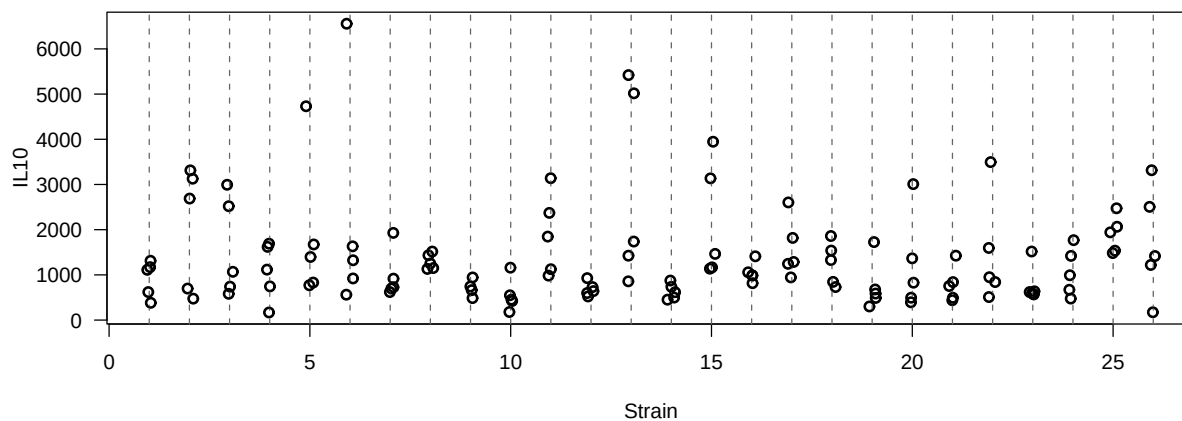


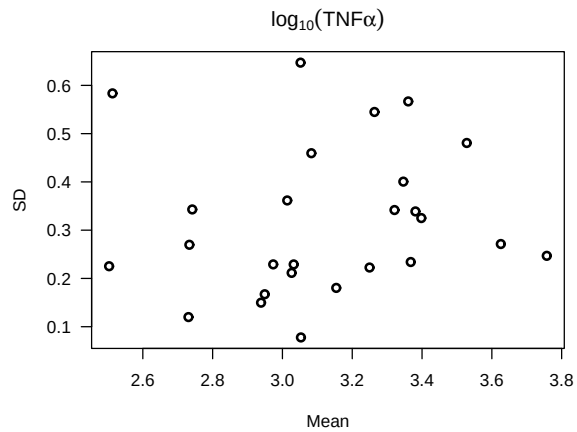
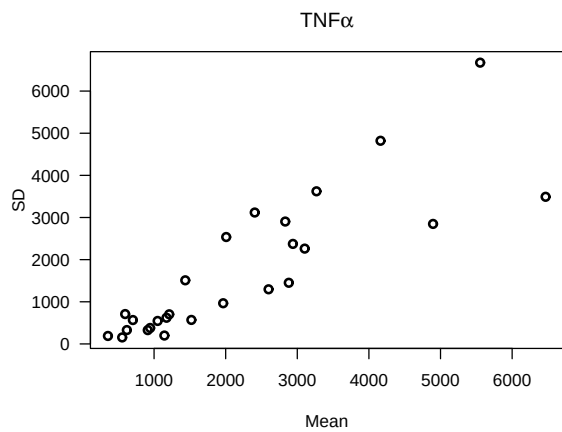
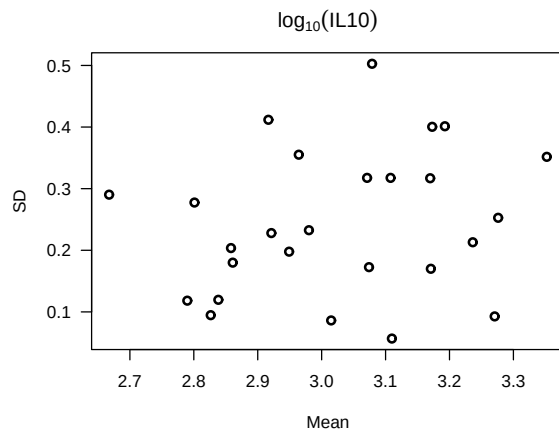
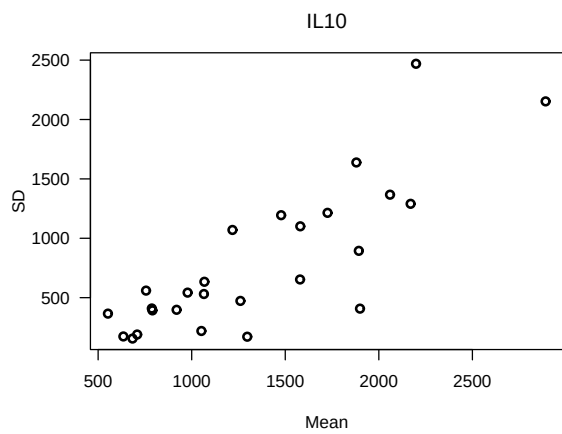
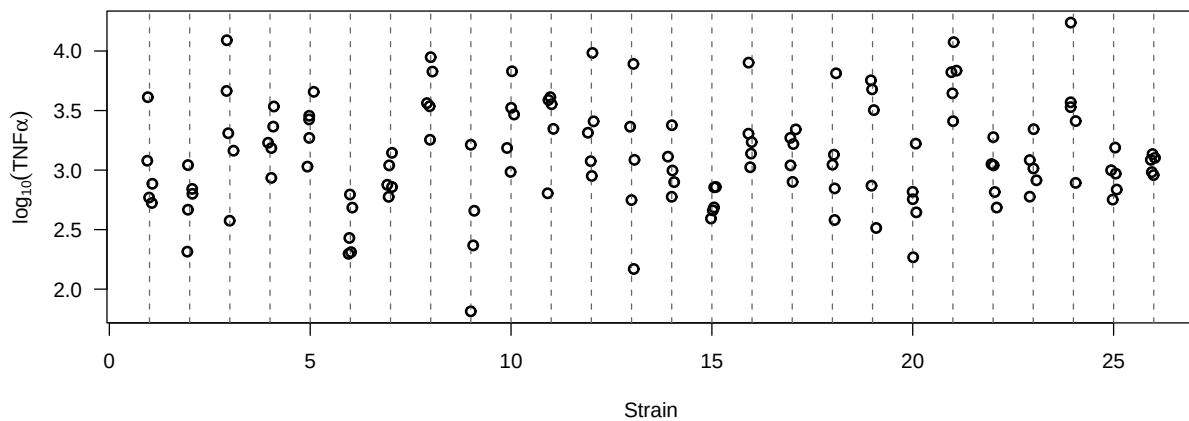
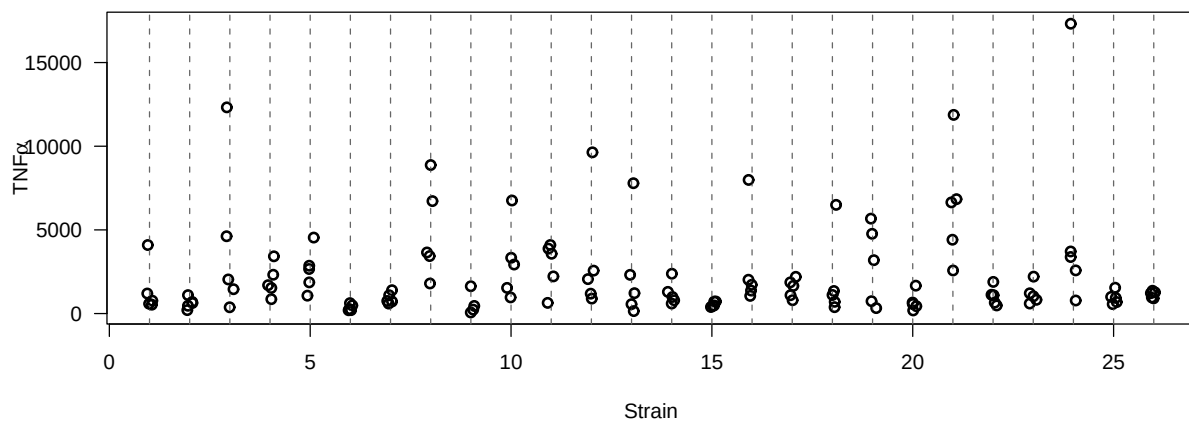
Why take logs?

- Statistics better behaved
- Stabilize SD (esp. if coefficient of variation constant)

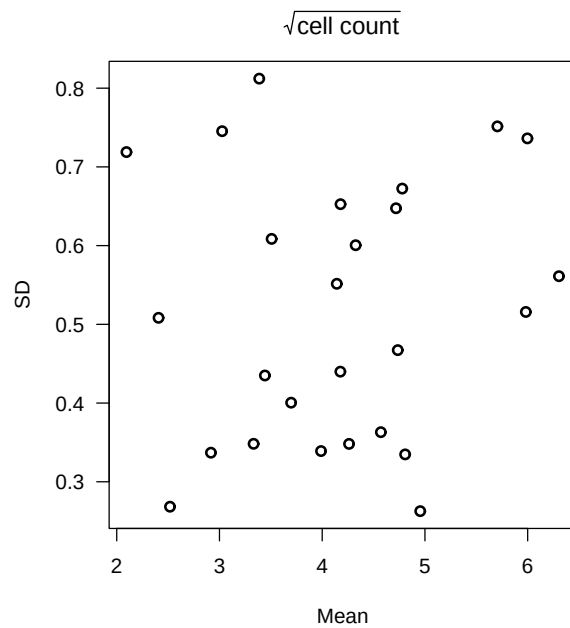
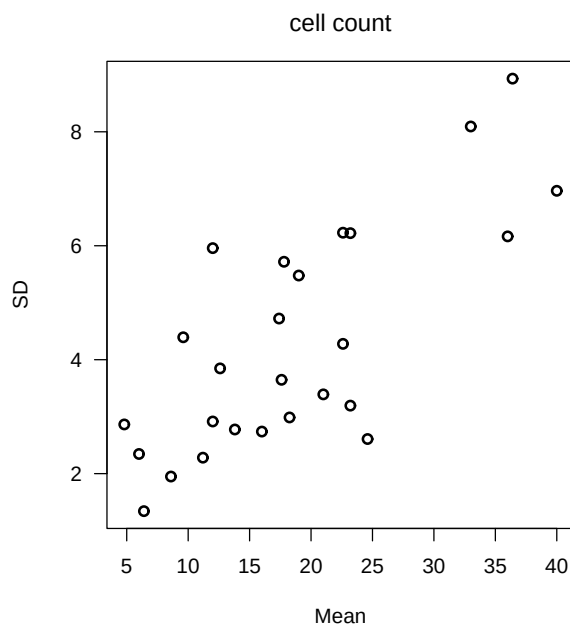
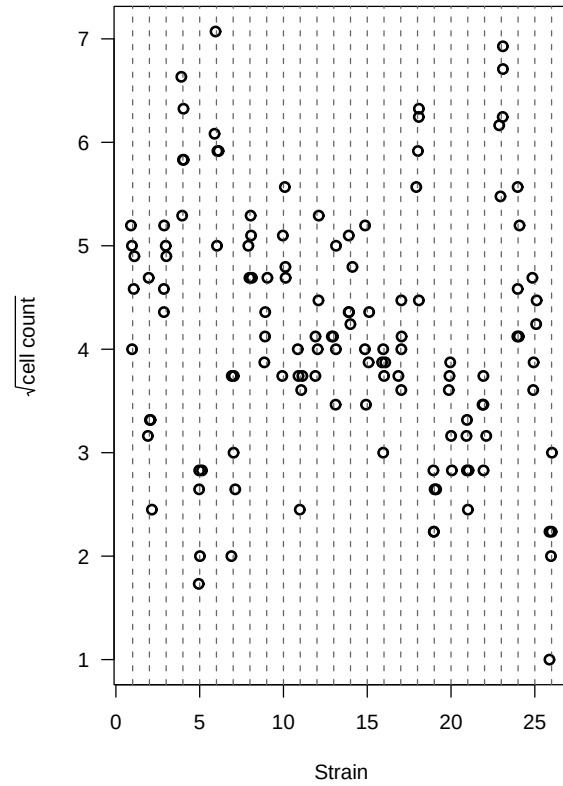
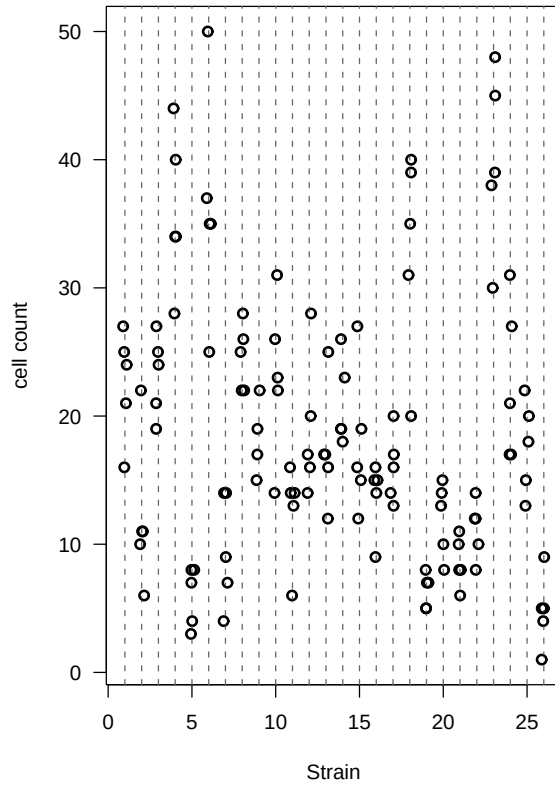
Note: $\text{mean}\{\log X\} = \log[\text{geo. mean}\{X\}]$

In the following: cytokine measurements (IL10 and $\text{TNF}\alpha$) in a set of recombinant inbred mouse strains.





Counts: take square root



Ratios: take logs

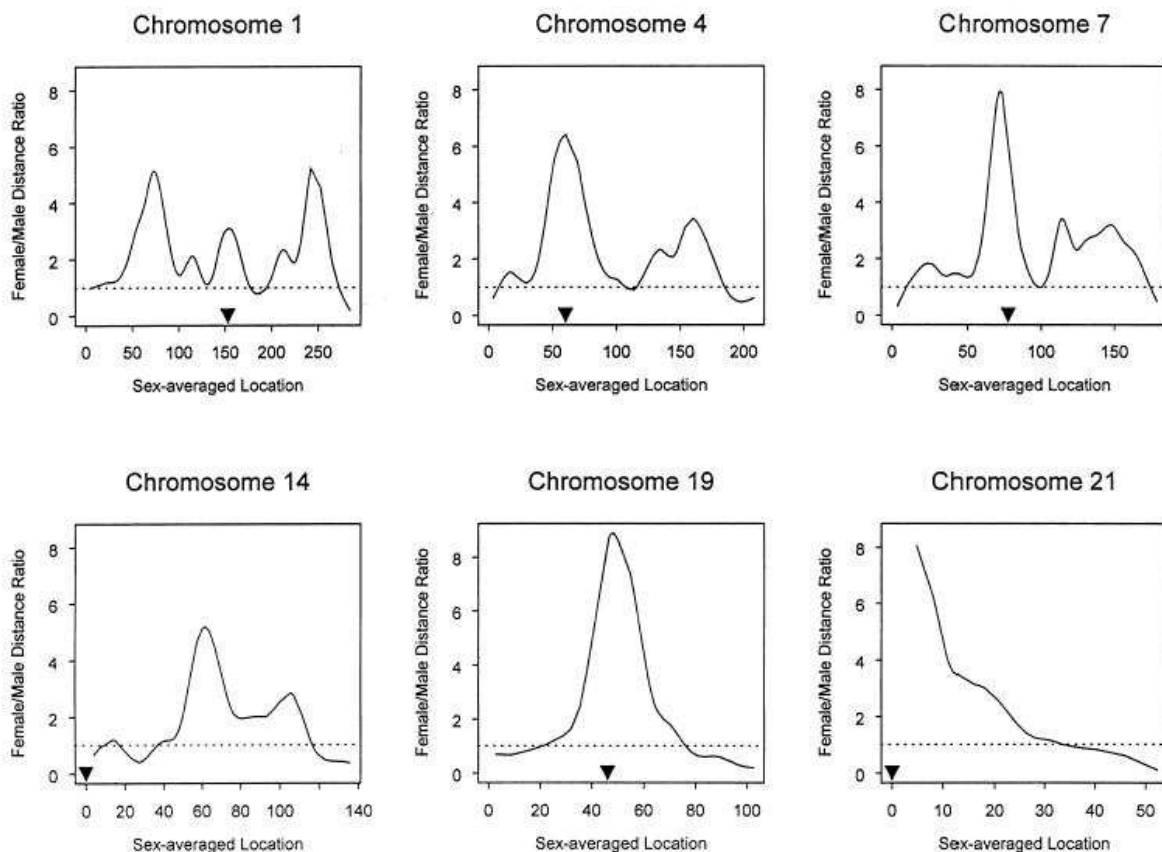
If you are interested in ratios of average responses, it might be better to look at the log ratios.

This insures that as much importance is given to ratios < 1 as to ratios > 1 .

Ratios: $(0,1)$ $(1,\infty)$

Log ratios: $(-\infty,0)$ $(0,\infty)$

The next figure should have been made on the log scale.



Outliers

“Outlier”: an odd-looking data point (far away from the others).
(This requires some sense of the scale.)

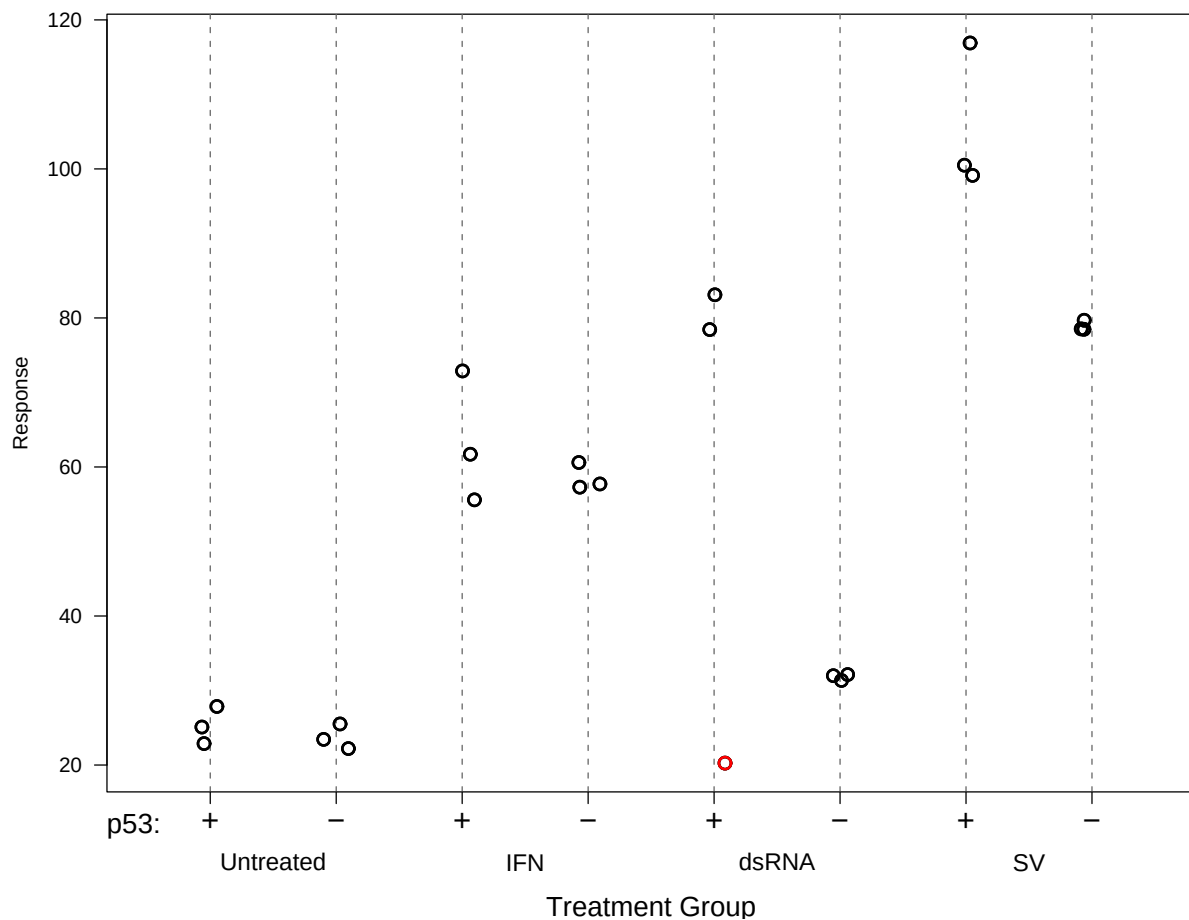
Q: Is it an error?

Q: Does it have undue influence on the results?

Example: (see data on next page)

With outlier: $P = 0.29$; 95% CI for mean diff. = $(-58, 116)$

Without outlier: $P = 0.029$; 95% CI for mean diff. = $(16, 82)$

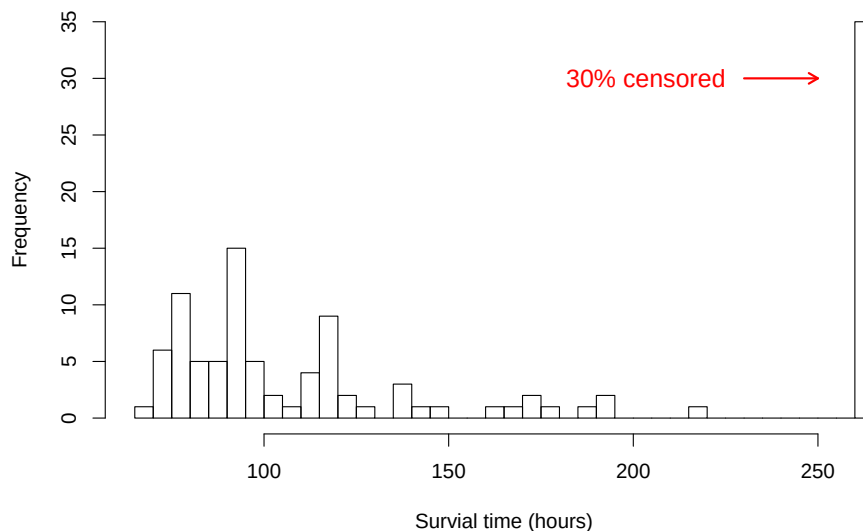


What to do with outliers

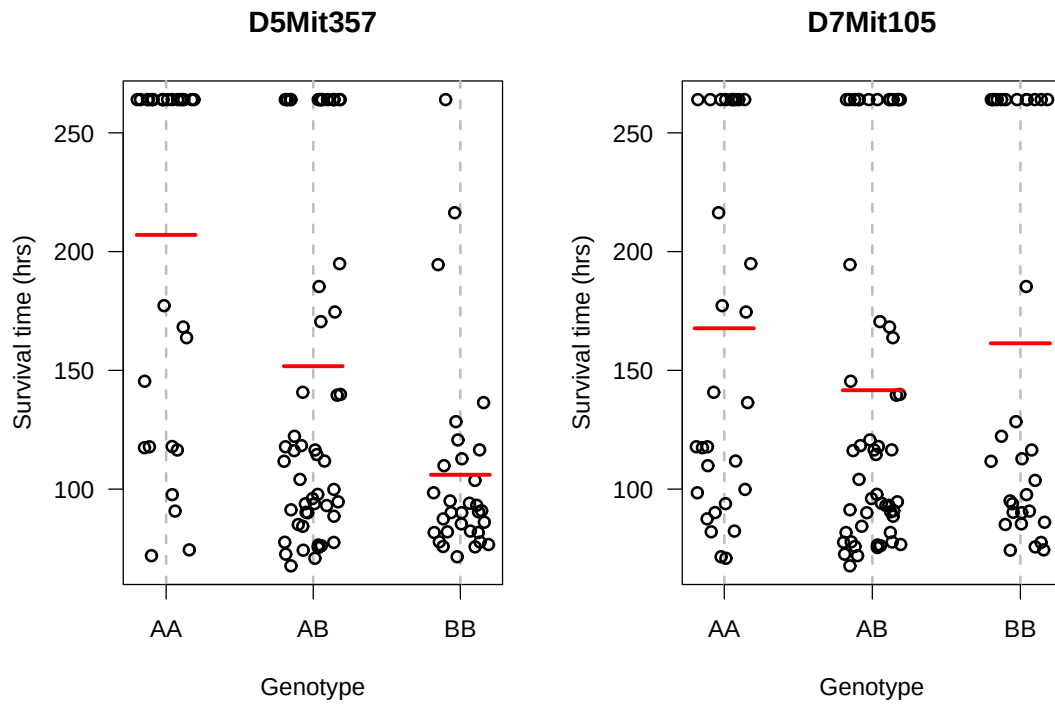
- Look at your data.
- Determine the cause.
- Determine the influence of the outlier.
- Consider a method of analysis that is **resistant** to the effects of outliers (aka **robust**). (This requires more than a tiny amount of data.)
- Delete outliers with great care, and **report it** if you do.

A spike in the distribution

Survival time in 120 intercross mice, following infection with *Listeria monocytogenes*



Phenotype by genotype



Phenotype by genotype

